
Review Article

A review of the biology and control of *Phlyctinus callosus* (Schönherr) (Coleoptera: Curculionidae), with special reference to biological control using entomopathogenic nematodes and fungi

B.E. Dlamini[§], P. Addison[§] & A.P. Malan^{*§}

Department of Conservation Ecology and Entomology, Stellenbosch University, Private Bag X1, Matieland, Stellenbosch, 7602 South Africa

The banded fruit weevil (BFW), *Phlyctinus callosus* (Schönherr) (Coleoptera: Curculionidae), is a polyphagous pest, feeding on a number of plant groups, like grasses, weeds, herbs, and woody plants. *Phlyctinus callosus* is indigenous to South Africa and has a limited distribution throughout the world. The weevil was only reported in the Western Cape, South Africa, New Zealand and Australia. Where the BFW occurs, it is a major insect pest of grapes and deciduous fruit, causing damage by feeding on the leaves and fruit, including the fruit stalks and shoots. The BFW adults, especially, cause fruit damage by means of feeding on the skin and the underlying flesh, which results in the formation of shallow lesions with phytosanitary implications. Other damage includes foliage damage, berry scarring and ringbarked stems, due to feeding activity. Control has mainly been achieved through the use of trunk barriers, which are effective in preventing the weevil from reaching the fruits, although such use is labour-intensive. Biocontrol agents, like entomopathogenic nematodes (EPNs) and entomopathogenic fungi (EPF), are preferred alternatives, because they have no negative effect on the environment, and because they leave no residues in the fruit intended for export. Using biological control agents in combination to control insect pests emphasises the importance of knowing the biology of the target insect pest, as well as that of the biocontrol agents, such as EPNs and EPF.

Key words: banded fruit weevil, deciduous fruit, grapevine, trunk barriers, biocontrol.

INTRODUCTION

The banded fruit weevil (BFW), *Phlyctinus callosus* (Schönherr) (Coleoptera: Curculionidae), which is one of the major insect pests of grapes and apples, is indigenous to South Africa (Lounsbury 1896; Barnes & Capatos 1989; Pringle *et al.* 2015). It is a phytosanitary pest, causing damage in grapevines, apple and nectarine orchards, particularly in the Western Cape Province (Pringle *et al.* 2015). BFW was recognised as a pest by Lounsbury in South Africa in 1896 (Kuschel 1972). Thereafter, it was first reported as a major pest of grapes in New Zealand in 1899 (Kuschel 1972), from where it spread to all the southern states in Australia. The BFW is the only curculionid that is considered to be a major pest of grapevines and deciduous fruits

in South Africa (Barnes & Pringle 1989). Furthermore, the BFW has the potential to inflict significant damage on several crops, as it has recently been found to be a serious pest of blueberries (Ferreira 2010; Pringle *et al.* 2015).

The BFW, which is a polyphagous pest, can feed on a number of plant groups, like grasses, weeds, herbs, and woody plants (Pringle *et al.* 2015). The type of host varies in the different countries in which the BFW has been recorded. In South Africa, the larvae can be found actively feeding on the roots of vines and on host weed species in the late winter/early spring (Fisher & Learmonth 2003; Pringle *et al.* 2015). The larvae tend to feed on the roots of a number of plants, while a few



*Author for correspondence. E-mail: apm@sun.ac.za

Received 3 April 2019. Accepted 6 June 2019

ISSN 1021-3589 [Print]; 2224-8854 [Online]
DOI: <https://doi.org/10.4001/003.027.0279>

African Entomology 27(2): 279–288 (2019)
©Entomological Society of Southern Africa

feed on weed roots in the mid-row, hence the presence of weeds influences the distribution and abundance of the BFW larvae (Pringle *et al.* 2015). Examples of broad-leaved weeds and ornamentals include the narrow-leaved ribwort, *Plantago lanceolata* (Plantaginaceae); hairy wild lettuce, *Hypochaeris radicata* (Asteraceae); bur clover, *Medicago* sp. (Fabaceae); sow thistle, *Sonchus oleraceus* (Asteraceae) and hen and chickens, *Chlorophytum comosum* (Asparagaceae) (Pringle *et al.* 2015).

The adults can feed on the stems and leaves of a number of weeds and grasses (Barnes 1989a,b; Barnes & Pringle 1989; Pringle *et al.* 2015). Damage caused by BFW adults in deciduous fruit and grapevine is normally on the leaves, fruits, fruit stalks and shoots. The adults also prefer to eat broad-leaved plants, just like in larvae, particularly those with fleshy leaves such as succulents (Pringle *et al.* 2015). Examples include the small mallow, *Malva parviflora* (Malvaceae); narrow-leaved ribwort; bur clover; sow thistle; purslane, *Portulaca oleracea* (Portulacaceae); sour fig, *Carpobrotus* spp. (Aizoaceae); *Coprosma rhamnoides* (Rubiaceae); chrysanthemum species (Asteraceae), and the succulent genera *Cotyledon* and *Crassula* (Crassulaceae) (Pringle *et al.* 2015).

The main aim of the current review was to consolidate the historical and currently available knowledge on BFW, and to provide information on the alternative control of BFW using biological control agents. The study emphasises the potential use of entomopathogenic nematodes (EPNs) and entomopathogenic fungi (EPF), and how the biologicals can be implemented in terms of an integrated pest management (IPM) system dedicated to the production of grapes and apples. Finally, the possibility of using such biologicals in combination, to obtain a synergistic effect for the control of BFW, is discussed.

PHLYCTINUS CALLOSUS AS PLANT PESTS

Identification

The larvae of BFW have an orange head that is exserted, their abdominal spiracles are inconspicuous and have a lightly sclerotised prothorax (May 1966). The lateral lobes of abdominal segments are broadly rounded with a protruding anal segment (May 1966). The larvae can easily be mistaken for the larvae of other weevils, including spotted vegetable weevil, *Desiantha diversipes*

(Pascoe), apple weevil, *Otiorhynchus cribricollis* (Gyllenhal), and sitona weevil, *Sitona discoideus* (Gyllenhal) (Fisher & Learmonth 2003). The last-instar larvae, on stoppage of feeding, make smooth-sided earthen cells, in which the prepupal stage changes into a soft-bodied pupal one. The BFW pupae are white, darkening as they mature; their eyes blacken, while the body becomes grey-brown, before adult emergence (Fisher & Learmonth 2003). As the adult BFW have fused elytra, they are flightless. They are greyish-brown, with a bulbous abdomen, which has the characteristic pale-white V-stripe that is prominent at the posterior margin.

Life cycle

The BFW is indigenous to the hot, dry summers and wet winters of the southwestern parts of South Africa (Lounsbury 1896; Annecke & Moran 1982; Pringle *et al.* 2015). The BFW can have one or two generations per annum, depending on the ground cover and the irrigation system used in the relevant vineyards or orchards during the dry summer conditions (Barnes 1989a). However, in its natural habitat, away from the irrigated orchards or vineyards, the BFW tends to have only one generation per year (Barnes 1989a; Pringle *et al.* 2015).

The eggs of the BFW are laid in batches of 20 or more on, or near, the soil surface in, or on, loose debris, foliage and organic litter (Barnes & Pringle 1989; Fisher & Learmonth 2003; Pringle *et al.* 2015). The BFW's growth is promoted by the maintenance of high ambient temperature (Barnes *et al.* 1994). In South Africa, the oviposition of the eggs, which occurs throughout the year, usually peaks from August to October (spring), December to January (summer), and March to May (autumn) (Barnes 1989a). From 6 to 15 days after hatching, the first-instar larvae burrow into the soil and feed on the roots of the host plants (Fisher & Learmonth 2003). Most larvae of the BFW undergo from 5 to 8 instars, but up to 11 instars can occur, depending on the prevailing weather conditions (Barnes 1989b; Pringle *et al.* 2015). The highest larval population normally occurs from June to October (second generation), and again from January to April (first generation) (Barnes 1989a; Pringle *et al.* 2015).

The prepupal stage constructs a chamber in the upper 100 mm of the soil, with the pupal stage lasting from 1 to 3 weeks. In South Africa, after

pupation, the BFW start emerging from the soil from October to December, in the case of the first generation, and, where the second generation is present, the adults start emerging from March to May (Barnes 1989a; Fisher & Learmonth 2003; Pringle *et al.* 2015). Fruit damage by the adult weevil becomes visible from December onwards, extending through to February (Fisher & Learmonth 2003). The adults, which feed during the night in the canopy, while seeking shelter during the day, can be controlled by means of trunk barriers, so as to prevent the weevils from reaching the fruit (Fisher & Learmonth 2003; Ferreira & Malan 2014; Pringle *et al.* 2015).

DAMAGE CAUSED BY *PHLYCTINUS CALLOSUS*

The damage to the fruit, which is caused by the adult weevils, is reported in relation to the grape berries and fruits of nectarines, apples, and other ornamental plants (Barnes 1989a, b; Barnes & Pringle 1989; Pringle *et al.* 2015). The adults cause fruit damage by means of feeding on the skin and the underlying flesh, which results in the formation of shallow lesions on the damaged fruit (Barnes 1989a; Fisher & Learmonth 2003; Ferreira & Malan 2014; Pringle *et al.* 2015).

Other damage that is caused by the BFW adults includes foliage damage, berry scarring and ring-barked stems, due to feeding activity (Barnes & Giliomee 1992; Fisher & Learmonth 2003; Pringle *et al.* 2015). Pringle *et al.* (2015) report that the adult weevil's damage to grapevines includes shot-holing of leaves and berries, leaf and stem notching, bud and shot-tip feeding, and scarring of the berries. The adults can feed on small green berries, causing them to drop prematurely (Pringle *et al.* 2015).

In South African apple orchards, the damage that is caused by the adults is of economic importance (Pringle *et al.* 2015). Just as with grape production, the BFW adults feed on the leaves, stems, and fruits. Adult damage includes leaf and stem notching, bud and shot-tip feeding, and scarring of the fruit (Pringle *et al.* 2015). Large populations of BFW can completely defoliate young fruit trees (Barnes 1989a, b). The scarring of the fruit results in unmarketable fruits and rejection of fruit consignments, due to the weevil being inadvertently packed in with the export fruits (De Villiers & Pringle 2007; Pringle *et al.* 2015).

The larvae of the BFW, in contrast, feed on the roots of vines, resulting in stunted growth, particularly in the young vines. The symptoms of water stress in young grapevines can be ascribed to the larvae of BFW feeding on the roots concerned (Fisher & Learmonth 2003). Root feeding in mature vines is, however, not damaging to the vines. It is important to control the larvae involved, because they indicate the extent of adult abundance and the severity of adult damage in the season to follow (Fisher & Learmonth 2003; Learmonth 2011). The BFW was reported by Barnes (1989a) to be causing sporadic damage to other deciduous fruits, like pears, plums and peaches.

CONTROL AND MANAGEMENT

Adult BFWs are flightless, so it is important to control all weeds and grasses growing among the host plants, as doing so reduces the number of larvae emerging from the soil (Pringle *et al.* 2015). A cultural control strategy involves cutting hanging host plant branches, which reduces the number of weevils that can reach the fruits concerned (Pringle *et al.* 2015). The use of trunk barriers is effective in preventing the BFW from reaching the fruits, but such use is labour-intensive, particularly in large orchards and vineyards (Barnes *et al.* 1994; Fisher & Learmonth 2003; Pringle *et al.* 2015). The trunk barriers commonly comprise low-density fibre (batting) strips, which are soaked in a synthetic pyrethroid, or a sticky substance, that is smeared on a plastic strip secured to the trunk (Pringle *et al.* 2015). The use of guinea fowl or chickens, which feed on the adults, larvae and pupae in the soil, in commercial plantings, has been reported to reduce the weevil's population below the economic threshold (Pringle *et al.* 2015). In Australia, where the insect was introduced, cultivating inter-rows in vineyards was reported to destroy a significant number of larvae and pupae (Pringle *et al.* 2015).

The use of bifenthrin (pyrethroid) in controlling black vine weevil, *Otiorhynchus sulcatus* (F.) (Coleoptera: Curculionidae), a close relative of the BFW, is less effective (Cowles 2003). Administering only low concentrations of bifenthrin allows some weevils to survive in certain parts of the plants (Cowles 2003). The currently used chemical control strategy consists of two early-season foliar esfenvalerate sprays, with one at 75 % petal fall and the other at 2 to 4 weeks after

petal fall (Vermeulen *et al.* 1990). A third insecticide spray is usually done mid-January, should it be required (Vermeulen *et al.* 1990). A number of apple growers normally apply the first BFW spray simultaneously with the second codling moth spray, 14 days after 75 % petal fall (Barnes & Giliomee 1992). The BFW is a sporadic pest and, where it is a problem, the population is kept in check by applying up to six sprays at 2- to 3-week intervals, so as to limit fruit damage (Barnes & Giliomee 1992). Although the synthetic pyrethroids applied can reduce the adult populations, they are not compatible with the IPM programmes currently used (Pringle *et al.* 2015).

The effects of conventional chemicals on the residues of food crops, on groundwater contamination, on non-target organisms, and on insect resistance to synthetic chemicals, have resulted in the development of alternative control measures, of which some include EPNs (Ferreira & Malan 2014) and EPF (Gillespie *et al.* 1988; Shahid *et al.* 2012). The BFW can develop resistance to pyrethroids, with the products being incompatible with the IPM programmes that are used in the orchards and vineyards in the Western Cape (Chagnon *et al.* 2015; Pringle *et al.* 2015). Fisher and Learmonth (2003) report that the foliar application of esfenvalerate to control the BFW results in secondary pest outbreaks, such as those of the long-tailed mealybug, *Pseudococcus longispinus* (Targioni Tozzetti) (Hemiptera: Pseudococcidae), the grape leaf rust mite, *Calepitrimerus vitis* (Nalepa) (Acari: Eriophyoidea) and the grapevine scale, *Parthenolecanium persicae* (Fabricius) (Hemiptera: Coccidae). The 'Kaolin particle film' technology can be used where fine-milled 95 % kaolin clay in aqueous suspension is sprayed onto the foliage of fruit trees, which results in the leaves and fruits being covered with a fine layer of kaolin (Pringle *et al.* 2015). The kaolin, which normally repels adult weevils, becomes difficult for the adults to walk on, thus obstructing feeding on the fruit (Pringle *et al.* 2015).

BIOLOGICAL CONTROL WITH NEMATODES AND FUNGI

Entomopathogenic nematodes

EPNs, which are from the order Rhabditida, consist of two families, namely Steinernematidae and Heterorhabditidae. EPNs belonging to the genera *Steinernema* and *Heterorhabditis* are obligate

parasites of insects. They kill their insect hosts with the help of their associated bacteria, *Xenorhabdus* and *Photorhabdus*, which are carried in the alimentary canal of the nematode (Kaya & Gaugler 1993; Shapiro-Ilan & Gaugler 2002; Shapiro-Ilan *et al.* 2003a). The free living infective juveniles (IJs) release the associated mutualistic bacterium from their gut after penetrating an insect gut and the mutualistic bacterium multiplies rapidly (Kaya & Gaugler 1993; Shapiro-Ilan & Gaugler 2002; Shapiro-Ilan *et al.* 2003a).

The bacteria serve as a source of nutrients for the nematodes, secreting antibiotics that inhibit any other competing microbes, and killing the insect host through the production of toxins, which leads to septicaemia. The most important role that the nematode plays in terms of the mutualistic relationship is to act as the bacterial vector, although they also contribute to the host insect's death by means of suppressing its immune system, and by means of producing toxins (Lu *et al.* 2017). However, the bacteria concerned cannot survive in nature in the absence of the nematodes, and they are not pathogenic when they are ingested by the host insect (Shapiro-Ilan & Gaugler 2002).

EPNs have been reported to be efficient in controlling a number of economically important insect pests, including the larvae of a number of weevil species, like the black vine weevil, the pecan weevil, *Curculio caryae* (Horn) (Coleoptera: Curculionidae) and the larvae of the diaprepes citrus root weevil, *Diaprepes abbreviatus* (L.) (Coleoptera: Curculionidae) (Shapiro-Ilan *et al.* 2002, 2003b). A number of insect pests in different habitats, particularly soil-dwelling and those in cryptic habitats, have been successfully controlled using EPNs (Campos-Herrera 2015). In such habitats, IJs tend to thrive easily, because they are protected from adverse environmental factors.

In a study by Ferreira & Malan (2014), different locally isolated EPNs were screened for their potential to control the BFW. Final instar larvae and adults were tested for their susceptibility against 400 infective juveniles (IJs) per insect after 4 days in 24-well bioassay trays. All the EPNs used in the study were able to control both larvae and adults, with the former being found to be more susceptible than the latter. *Heterorhabditis zealandica* Poinar, which was found to be the most effective isolate, was used to study the impact of vertical movements of the EPNs in sand and sandy

loam soil. The sandy loam soil was found to give higher percentage mortality than did the sandy soil.

Dlamini *et al.* (2019) screened different EPNs for virulence against different stages of BFW, followed by the application of *Steinernema yirgalemense* Nguyen, Tesfamariam, Gozel, Gaugler & Adams, in a field trial at different concentrations. Three EPNs, *S. yirgalemense*, *Heterorhabditis noenieputensis* Malan, Knoetze & Tiedt and an exotic isolate of *Steinernema feltiae* Filipjev gave good mortality of BFW larvae at 100 IJs per insect, after 2 days. *Heterorhabditis indica* Poinar, Karunakar & David (70 %) and *Heterorhabditis baujardi* Phan, Subbotin, Nguyen & Moens (67 %) gave good mortality for BFW pupae, compared with that provided by *H. noenieputensis* (55 %). Higher mortality of BFW adults was reported with *H. indica* (95 %) and *S. yirgalemense* (94 %), compared to with the three other EPNs tested. In the field trials, relatively high mortality of BFW was reported when *S. yirgalemense* were exposed to BFW larvae at 20 and 40 IJs/cm², resulting in 69 % and 78 % control after 48 h of exposure, respectively.

EPNs were able to control 35 % of *C. caryae* larvae in field trials when *Heterorhabditis bacteriophora* Poinar, *Steinernema carpocapsae* (Weiser, 1955) Wouts, Mráček, Gerdin & Bedding, or *S. feltiae*, were used (Nyczepir *et al.* 1992; Smith *et al.* 1993; Shapiro-Ilan *et al.* 2004). Laboratory studies on the virulence of nine species of EPNs resulted in less than 60 % mortality, when applied at a concentration of 40 IJs/cm² (Shapiro-Ilan 2001a, b). In a study conducted by Kakouli-Duarte *et al.* (1997), in which field trials were conducted on *O. sulcatus* larvae in strawberry plants, using *S. carpocapsae* and *Heterorhabditis megidis*, with the former treatment, administered in late summer, resulting in 49.5 % reduction of early instar *O. sulcatus* larvae, while late spring application produced 65 % control of late instar larvae. When *H. megidis* was applied during the spring, 26 % mortality of late instar larvae was obtained. The above was after the EPNs were both applied using drip irrigation across raised strawberry beds. In other trials by Haukeland & Lola-Luz (2010), *Heterorhabditis megidis* resulted in good control of *O. sulcatus*, as long as the temperature lasted above 10 °C. In another study, the larvae of *D. abbreviatus*, after being exposed to 12 strains of *Steinernema riobrave* Cabanillas, Poinar & Raulston 1994 resulted in larval mortality that ranged from 58 % to 84 %

when commercial and mixed strains were used, with the latter resulting in 46 % more control than did the former (Stuart *et al.* 2004).

Booth *et al.* (2002) report that *S. carpocapsae* and *Steinernema glaseri* Wouts, Mráček, Gerdin & Bedding when applied at 30 IJs/cm² resulted in 96 % and 100 % reduced abundance of *O. sulcatus*, respectively. A bait bioassay using wax moth, *Galleria mellonella* (L.) (Lepidoptera: Pyralidae), showed that *S. carpocapsae* and *S. glaseri* persisted for 0.48 % and 0.04 % infested wax moth per applied IJ at 30 days after treatment, with 60 days after treatment as a longer interval. *Otiiorhynchus sulcatus* and *Otiiorhynchus ovatus* Linnaeus (Coleoptera: Curculionidae) were reduced in number when *S. carpocapsae* and *Heterorhabditis marelatus* Liu & Berry, 1996 were applied to strawberry sites, compared with what was achieved with the untreated control (Booth *et al.* 2002). When the same EPNs were applied at the same rates at other sites, only pupal black vine weevils were significantly affected. In laboratory bioassays involving the pupal strawberry root weevil, *O. ovatus*, *S. carpocapsae* and *H. bacteriophora* resulted in 62 % and 51 % mortality, respectively (Simser & Roberts 1994). After the EPNs were applied post-harvest, limited movement/distribution of the EPNs occurred from the soil surface to 40 mm soil depth after 336 h (Simser & Roberts 1994). After increasing the spray pressure, the population of *O. ovatus* reduced, from 26.7 to 0.6 live larvae/sample and from 16.2 to 1.8 live larvae/sample after 240 h, after the application of *S. carpocapsae* and *H. bacteriophora*, respectively (Simser & Roberts 1994). The bioassay trial resulted in 74 % and 69 % mortality for *S. carpocapsae* and *H. bacteriophora*, respectively.

Entomopathogenic fungi

EPFs, which are insect pest parasites that show disease symptoms in the host insects, are important insect pathogens that significantly reduce the number of host insects and arthropod populations in nature (Shahid *et al.* 2012). Although the parasites live in association with the host insect, they benefit at its expense (Shahid *et al.* 2012). The causation of lethal infections by EPF serves as an ecosystem function, in terms of regulating the population of insects and mites by means of epizootics. In addition, EPF are host-specific, with the risk of attacking such non-target organisms such as beneficial insects. Also, EPF can infect an extensive range of economic insect pests, including

lepidopterous larvae, aphids, beetles, and thrips (Shahid *et al.* 2012).

Approximately 750 species of EPF from 85 different genera are currently known (Shahid *et al.* 2012). EPF are good biocontrol agents, because they act on contact, and they do not require ingestion. In addition, they are able to be easily mass-produced and formulated (Shahid *et al.* 2012). EPFs are grouped into the following divisions: Zygomycota; Ascomycota and Deuteromycota; and Chytridiomycota and Oomycota (Shahid *et al.* 2012). The members of a group of over 750 species from 85 genera, as found in the different classes of fungi, can parasitise almost all insect groups (Shahid *et al.* 2012). Deshpande (2008) reported that most groups containing EPF include genera such as *Metarhizium*, *Beauveria*, *Verticillium*, *Nomuraea*, *Entomophthora*, and *Neozygites*. Notably, however, fungi can infect other soil arthropod species than the insect pests of cultivated crops (Zimmermann 1993; Shahid *et al.* 2012). EPFs, which are reported to be infective to all the life stages of their insect hosts, are found in almost all soils worldwide.

A key EPF that is present in soils throughout the world is *Metarhizium anisopliae* (Metschnikoff) (Hypocreales: Clavicipitaceae). The fungus, which was first used as a biological control agent in the 1880s (Shahid *et al.* 2012), can infect and successfully control four insect orders, namely Coleoptera, Isoptera, Hemiptera, and Orthoptera (Zimmermann 1993; Shahid *et al.* 2012). A number of spores, or mycelia formulations, of *M. anisopliae* can be applied. After a fungal infection is achieved, the spores and vegetative cells produced from the infected insect tend to spread to other populations of insects (Zimmermann 1993; Shahid *et al.* 2012). Gottwald & Tedders (1983) report that, when *B. bassiana* and *M. anisopliae* were used in field and greenhouse trials to control populations of *O. sulcatus*, 72 % and 50 % mortality of the adults resulted, respectively. In the same study, when a single adult contaminated with the EPF was caged with untreated adult weevils, 44.4 % and 27.8 % mortality was reported for *B. bassiana* and *M. anisopliae*, respectively. After *B. bassiana* and *M. anisopliae* were soil-applied, 30 % and 6 % larval mortality was achieved, respectively (Gottwald & Tedders 1983). In another study, undertaken by Dembilio *et al.* (2010), the potential of *B. bassiana* originally isolated from a naturally infected red palm weevil, *Rhynchophorus*

ferrugineus Oliver (Coleoptera: Curculionidae), pupa showed that the EPF could infect the eggs, larvae and adults of *R. ferrugineus*. The adults of *R. ferrugineus* inoculated with the EPF were able to transmit the disease to untreated adults. The rate of transmission from male-to-female and from female-to-male was 55 % and 60 %, respectively. A reduced fecundity of up to 63 %, and reduced egg hatching of up to 33 % after the weevil was treated with *B. bassiana*, was observed (Dembilio *et al.* 2010).

In South Africa, the fungus, *Metarhizium acridum* strain IMI 330189, was registered as part of the LUBILOSA (Lutte Biologique contre les Locustes et Sauteriaux) project (Lomer *et al.* 2001) against locusts and grasshoppers (Bateman *et al.* 1994; Price *et al.* 1997), leading to the registration of Green Muscle® in 1998 (Müller 2000; Hatting *et al.* 2018). A number of species are categorised under the *M. anisopliae sensu lato* species complex, but members of the 'PARB' clade, including *Metarhizium pinghaense*, *M. anisopliae*, *M. robertsii*, and *M. brunneum*, are important as EPFs (Bischoff *et al.* 2009; Rehner & Kepler 2017; Hatting *et al.* 2018). *Metarhizium anisopliae* and *M. brunneum* have successfully been commercialised in a number of countries, with *M. anisopliae*, isolate ICIPE69, being registered against thrips, whiteflies and snout beetles in South Africa (De Faria & Wraight 2007; Zimmermann 2007a; Ekesi *et al.* 2011; Rehner & Kepler 2017; Hatting *et al.* 2018).

Of the 18 known *B. bassiana* species, only four have been reported in South Africa, including *Beauveria bassiana* (Bals.-Criv.) Vuill. (1912), *Beauveria brongniartii* (Saccardo) Petch (De Hoog 1972), *Beauveria caledonica* Bissett & Widden, and *Beauveria pseudobassiana* (Bals.) Vuill. (Rong & Grobbelaar 1998; Morar-Bhana *et al.* 2011; Goble *et al.* 2012; Abaajeh 2015; De Beer *et al.* 2017; Hatting *et al.* 2018). A total of five *B. bassiana*-containing products are registered in South Africa: BB Plus WP®, which is registered to control aphids and spider mites; BB Weevil EC®, which is registered to control curculionids; Beauvitech® WP, which is registered against whitefly; Broadband®, which is registered against red spider mite, diamond-back moth, potato tuber moth, whitefly, thrips, stinkbug, false codling moth, and red scale; and Eco-Bb®, which is registered against red spider mite, whitefly, fall army worm, false codling moth, and tomato leaf miner (Zimmermann 2007b; Hatting *et al.* 2018).

COMBINED USE OF NEMATODES AND FUNGI

In using organisms as control agents, the major differences in their biology should be taken into account when considering their use to control insect pests. EPNs are multicellular organisms, with specific oxygen and moisture requirements. In contrast, EPF are single cellular organisms, and the environmental factors do not play a crucial role as in the case of EPNs, giving them an advantage in terms of mass culture and storage (Emelianoff *et al.* 2008; Shahid *et al.* 2012). A major advantage in using EPNs as a control agent is that they have the ability to detect their host *via* amphids or sensory organs, as they move actively towards the target host, hiding in cryptic habitats, such as in the soil, where insect larvae can also hide in cocoons, as prepupae or pupae (Emelianoff *et al.* 2008; Shahid *et al.* 2012). In the case of EPF, spores are stationary and can only spread in the soil during their application with water. However, moisture could be a negative in the case of the use of fungi, as it triggers germination in the spores, without necessarily being attached to an insect. Above ground, EPF also have a slight advantage, as they can dry out completely on the host plant, and yet still become attached to a passing insect (Emelianoff *et al.* 2008; Shahid *et al.* 2012). Nematodes, however, require a water film to move above-ground, and tend to die as soon as the water evaporates. Knowledge of the biological advantages and disadvantages of the biological control agent involved is important in terms of the combined use of nematodes and fungi, with the aim of enhanced control.

Apart from the evidence that entomopathogens are effective in controlling weevils, the combined use of both microbial control agents has additional advantages to their individual use. Such benefits include their potential to infect the insects in cryptic habitats, to reduce the pesticide residues on food, to facilitate mass production, and not to endanger the safety of humans and a number of non-target organisms (Emelianoff *et al.* 2008; Malan *et al.* 2011; Shahid *et al.* 2012). Shahid *et al.* (2012) who report that entomopathogens manage the population of insects and mites in nature by means of causing lethal infections, which tend to result in temporary outbreaks of disease. The improved efficacy of the combined use of biocontrol agents comes from the synergistic and additive effects of the biocontrol agents involved (Koppenhöfer & Kaya 1997; Koppenhöfer *et al.* 1999; Ansari *et al.* 2008).

Shapiro-Ilan *et al.* (2003a) report that both *Heterorhabditis* and *Steinernema* spp. and fungi (*B. bassiana* and *M. anisopliae*) are pathogenic to a number of insect groups, including coleopteran pests. In addition, Ansari *et al.* (2008) report that the black vine weevil can easily be controlled by a combination of EPNs and EPFs. The heterorhabditid and steinernematid EPNs, when combined with *M. anisopliae*, have been shown to control the black vine weevil efficiently. In addition, the ability of EPNs to respond to such host volatiles as faeces and carbon dioxide facilitates them finding the immature stages of the BFW in the soil (Grewal *et al.* 1997; Le Vieux & Malan 2015).

In another study by Ansari *et al.* (2008), the application of *M. anisopliae*, followed by the application of either *H. bacteriophora*, *S. feltiae* or *S. kraussei* after 1 or 2 weeks, provided 100 % control of third-instar *O. sulcatus*. A synergistic effect was observed when the EPNs and *M. anisopliae* were used to control *O. sulcatus* in potted winter creeper, *Euonymus fortune* (Celastrales: Celastraceae), under greenhouse conditions (Ansari *et al.* 2008; Lacey & Georgis 2012).

CONCLUSION

Research on *P. callosus* is scant, as the insect is currently only reported from South Africa, New Zealand and Australia. *Phlyctinus callosus* is also an important quarantine organism, because of its limited distribution. The BFW is regarded as a serious pest of grapevine in all the countries from which it has been reported. Their sporadic nature adds to its difficulty to control, especially in terms of an integrated pest management system. Biological control agents can play a major future role in the control of hot spots or sporadic flushes of *P. callosus* in a localised area, without interfering with, and undermining, the local pest control programmes. EPNs, as a biological control agent, are highly virulent, causing high mortality to insect pests within 2 days (Simões, & Rosa 1999). Ferreira & Malan (2014) and Dlamini *et al.* (2019) have already reported that local South African EPNs can be effectively used as biological control agents against all the different life stages of the BFW, not only in the laboratory, but also in field applications. Previous studies also demonstrated promising results with the use of *Beauveria* and *Metarhizium* for the control of *O. sulcatus* within 21 days (Grewal *et al.* 1997; Ansari *et al.*

2008). Furthermore, the combination of both the EPNs and EPF results in high mortality of the BFW larvae, as the combined use of microbial control agents has both synergistic and additive effects (Koppenhöfer & Kaya 1997; Koppenhöfer *et al.* 1999; Ansari *et al.* 2008). The current study reviews the biology of BFW and indicates the importance of using biological control agents for the control of *P. callosus*, with more emphasis on coming up with good IPM and soil health programmes.

REFERENCES

- ABAAJEH, A.R. & NCHU, F. 2015. Isolation and pathogenicity of some South African entomopathogenic fungi (Ascomycota) against eggs and larvae of *Cydia pomonella* (Lepidoptera: Tortricidae). *Biocontrol Science and Technology* **25**: 828–842.
DOI: <https://doi.org/10.1080/09583157.2015.1019831>
- ANNECKE, D.P. & MORAN, V.C. 1982. *Insects and Mites of Cultivated Plants in South Africa*. Butterworths, Durban/Pretoria, South Africa.
- ANSARI, M.A., SHAH, F.A. & BUTT, T.M. 2008. Combined use of entomopathogenic nematodes and *Metarhizium anisopliae* as a new approach for black vine weevil, *Otiorhynchus salcatus*, control. *Entomologia Experimentalis et Applicata* **129**: 340–347.
DOI: <https://doi.org/10.1111/j.1570-7458.2008.00783.x>
- BARNES, B.N. 1989a. Different life and seasonal cycles of BFW, *Phlyctinus callosus* (Coleoptera: Curculionidae), in apple orchards in the South-Western Cape. *Phytophylactica* **21**: 147–157.
- BARNES, B.N. 1989b. Embryonic and immature stages of *Phlyctinus callosus* Boh. (Coleoptera, Curculionidae) – aspects of biology and behaviour with respect to control in deciduous fruit orchards. *Journal of the Entomological Society of Southern Africa* **52**: 165–178.
- BARNES, B.N. & CAPATOS, D. 1989. Evidence for an aggregation pheromone in adult frass of BFW, *Phlyctinus callosus* (Schoenherr) (Col., Curculionidae). *Journal of Applied Entomology* **108**: 512–518.
DOI: <http://dx.doi.org/10.1111/2Fj.1439-0418.1989.tb00485.x>
- BARNES, B.N. & PRINGLE, K.L. 1989. Oviposition by BFW, *Phlyctinus callosus* (Schoenherr) (Coleoptera: Curculionidae) in deciduous fruit orchards in South Africa. *Bulletin of Entomological Research* **79**: 31–40.
DOI: <http://dx.doi.org/10.1017/2FS000748530001854X>
- BARNES, B.N. & GILIOME, J.H. 1992. Fruit-feeding behaviour of banded fruit weevil, *Phlyctinus callosus* (Schönherr) (Col., Curculionidae), in apple orchards. *Journal of Applied Entomology* **113**: 407–415.
DOI: [10.1111/j.1439-0418.1992.tb00681.x](https://doi.org/10.1111/j.1439-0418.1992.tb00681.x)
- BARNES, B.N., KNIPE, M.C. & CALITZ, F.J. 1994. Trunk barriers provide effective control of banded fruit weevil on apples and nectarines. *Deciduous Fruit Grower* **44**: 322–327.
- BATEMAN, R.P., PRICE, R.E., MULLER, E.T. & BROWN, H.D. 1994. Controlling brown locust hopper bands in South Africa with a mycopesticide spray. In: *Proceedings of the Brighton Crop Protection Conference – Pests and Diseases* **6A-3**: 609–616.
- BISCHOFF, J.F., REHNER, S.A. & HUMBER, R.A. 2009. A multilocus phylogeny of the *Metarhizium anisopliae* lineage. *Mycologia* **101**: 512–530.
DOI: <https://doi.org/10.3852/07-202>
- BOOTH, S.R., TANIGOSHI, L.K., & SHANKS, C.H., J.R. 2002. Evaluation of entomopathogenic nematodes to manage root weevil larvae in Washington State cranberry, strawberry, and red raspberry. *Environmental Entomology* **31**: 895–902.
DOI: <https://doi.org/10.1603/0046-225X-31.5.895>
- CAMPOS-HERRERA, R. 2015. *Nematode Pathogenesis of Insects and Other Pests; Ecology and Applied Technologies for Sustainable Plant and Crop Protection*. Springer International Publishing, Neuchâtel, Switzerland.
- CHAGNON, M., KREUTZWEISER, D., MITCHELL, E.A., MORRISSEY, C.A., NOOME, D.A. & VAN DER SLUIJS, J.P. 2015. Risks of large-scale use of systemic insecticides to ecosystem functioning and services. *Environmental Science and Pollution Research* **22**: 119–134.
DOI: <https://doi.org/10.1007/s11356-014-3277-x>
- COWLES, R.S. 2003. Practical black vine weevil management. *Journal ARS* **57**: 219–223.
- DE BEER, Z.W., DE BRUIJN, H., NEL, W.J. & SLIPPERS, B. 2017. A survey of entomopathogenic *Beauveria* spp. in South African plantations. In: *Proceedings of the 2017 Combined Congress of the Entomological and Zoological Societies of Southern Africa*, CSIR International Convention Centre, Pretoria, South Africa, 3–7 July 2017, p. 422.
- DE FARIA, M.R. & WRIGHT, S.P. 2007. Mycoinsecticides and mycoacaricides: a comprehensive list with worldwide coverage and international classification of formulation types. *Biological Control* **43**: 237–256.
DOI: <http://dx.doi.org/10.1016/j.biocontrol.2007.08.001>
- DEMBILIO, Ó., QUESADA-MORAGA, E., SANTIAGO-ÁLVAREZ, C. & JACAS, J.A. 2010. Potential of an indigenous strain of the entomopathogenic fungus *Beauveria bassiana* as a biological control agent against the red palm weevil, *Rhynchophorus ferrugineus*. *Journal of Invertebrate Pathology* **104**(3): 214–221.
DOI: <https://doi.org/10.1016/j.jip.2010.04.006>
- DESHPANDE, M.V. 2008. Mycopesticide production by

ACKNOWLEDGEMENTS

We would like to thank HortGro Pome, the South African Table Grape Industry, Winetech, and the Technology and Human Resources for Industry Programme (THRIP: TP14062571871) for funding of the project.

*ORCID iDs

- B.E. Dlamini:  orcid.org/0000-0002-5289-7433
P. Addison:  orcid.org/0000-0002-8227-339X
A.P. Malan:  orcid.org/0000-0002-9257-0312

- fermentation: potential and challenges. *Critical Reviews in Microbiology* **25**(3): 229–243.
DOI: <https://doi.org/10.1080/10408419991299220>
- DE VILLIERS, M. & PRINGLE, K.L. 2007. Seasonal occurrence of vine pests in commercially treated vineyards in the Hex River Valley in the Western Cape Province, South Africa. *African Entomology* **15**: 241–260.
DOI: <https://doi.org/10.4001/1021-3589-15.2.241>
- DLAMINI, B.E., MALAN, A.P. & ADDISON, P. 2019. Control of the banded fruit weevil *Phlyctinus callosus* (Coleoptera: Curculionidae) using entomopathogenic nematodes. *Austral Entomology*.
DOI: <https://doi.org/10.1111/aen.12386>
- EKESI, S., CHABI-OLAYE, A., SUBRAMANIAN, S. & BORGEMEISTER, C. 2011. Horticultural pest management and the African economy: successes, challenges and opportunities in a changing global environment. *Acta Horticulturae* **911**: 165–183.
DOI: [10.17660/ActaHortic.2011.911.17](https://doi.org/10.17660/ActaHortic.2011.911.17)
- EMELIANOFF, V., CHAPUIS, E., LE BRUN, N., CHIRAL, M., MOULLA, C. & FERDY, J.B. 2008. A survival-reproduction trade-off in entomopathogenic nematodes mediated by their bacterial symbionts. *Evolution* **62**: 932–942.
- FERREIRA, T. 2010. Rearing of the BFW, *Phlyctinus callosus* (Schönherr) (Coleoptera: Curculionidae) and control with entomopathogenic nematodes. M.Sc. thesis, Stellenbosch University, Stellenbosch, South Africa.
- FERREIRA, T. & MALAN, A.P. 2014. Potential of entomopathogenic nematodes for the control of the BFW, *Phlyctinus callosus* (Schönherr) (Coleoptera: Curculionidae). *Journal of Helminthology* **88**: 293–301.
DOI: <https://doi.org/10.1017/S0022149X13000175>
- FISHER, D. & LEARMONTH, S. 2003. Garden weevil in vineyards. *Western Australia Department of Agriculture. Farmnote No. 60/2003*.
- GILLESPIE, J.P., BATEMAN, R. & CHARNLEY, A.K. 1988. Role of cuticle-degrading proteases in the virulence of *Metarhizium* spp. for the desert locust, *Schistocerca gregaria*. *Journal of Invertebrate Pathology* **71**: 128–137.
DOI: [10.1006/jipa.1997.4733](https://doi.org/10.1006/jipa.1997.4733)
- GOBLE, T.A., COSTET, L., ROBENE, I., NIBOUCHE, S., RUTHERFORD, R.S., CONLONG, D.E. & HILL, M.P. 2012. *Beauveria brongniartii* on white grubs attacking sugarcane in South Africa. *Journal of Invertebrate Pathology* **111**: 225–236.
DOI: <https://doi.org/10.1016/j.jip.2012.08.010>
- GOTTWALD, T.R. & TEDDERS, W.L. 1983. Suppression of pecan weevil (Coleoptera: Curculionidae) populations with entomopathogenic fungi. *Environmental Entomology* **12**(2): 471–474.
DOI: <https://doi.org/10.1093/ee/12.2.471>
- GREWAL, P.S., LEWIS, E.E. & GAUGLER, R. 1997. Response of infective stage parasites (Nematoda: Steinernematidae) to volatile cues from infected hosts. *Journal of Chemical Ecology* **23**: 503–515.
DOI: <https://doi.org/10.1023/B:JOEC.0000006374.95624.7e>
- HATTING, J.L., MOORE, S.D. & MALAN, A.P. 2018. Microbial control of phytophagous invertebrate pests in South Africa: current status and future prospects. *Journal of Invertebrate Pathology*.
DOI: [10.1016/j.jip.2018.02.004](https://doi.org/10.1016/j.jip.2018.02.004)
- HAUKELAND, S. & LOLA-LUZ, T. 2010. Efficacy of the entomopathogenic nematodes *Steinernema kraussei* and *Heterorhabditis megidis* against the black vine weevil *Otiiorhynchus sulcatus* in open field-grown strawberry plants. *Agricultural and Forest Entomology* **12**: 363–369.
DOI: [10.1111/j.1461-9563.2010.00497.x](https://doi.org/10.1111/j.1461-9563.2010.00497.x)
- KAKOULI-DUARTE, T., LABUSCHAGNE, L., & HAGUE, N.G.M. 1997. Biological control of the black vine weevil, *Otiiorhynchus sulcatus* (Coleoptera: Curculionidae) with entomopathogenic nematodes (Nematoda: Rhabditida). *Annals of Applied Biology* **131**: 11–27.
DOI: <https://doi.org/10.1111/j.1744-7348.1997.tb05393.x>
- KAYA, H.K. & GAUGLER, R. 1993. Entomopathogenic nematodes. *Annual Review of Entomology* **38**: 181–206.
DOI: <https://doi.org/10.1146/annurev.en.38.010193.001145>
- KOPPENHÖFER, A.M. & KAYA, H.K. 1997. Additive and synergistic interactions between entomopathogenic nematodes and *Bacillus thuringiensis* for scarab grub control. *Biological Control* **8**: 131–137.
DOI: <https://doi.org/10.1006/bcon.1996.0498>
- KOPPENHÖFER, A.M., CHOO, H.Y., KAYA, H.K., LEE, D.W. & GELERNTER, W.D. 1999. Increased field and greenhouse efficacy against scarab grubs with a combination of an entomopathogenic nematode and *Bacillus thuringiensis*. *Biological Control* **14**: 37–44.
DOI: <https://doi.org/10.1006/bcon.1998.0663>
- KUSCHEL, G. 1972. The foreign Curculionoidae established in New Zealand (Insecta: Coleoptera). *New Zealand Journal of Science* **15**(3): 273–289.
- LACEY, L.A. & GEORGIS, R. 2012. Entomopathogenic nematodes for control of insect pests above and below ground with comments on commercial production. *Journal of Nematology* **44**(2): 218–225.
- LEARMONTH, S. 2011. Improved management of garden weevil in WA – a warning service and assessing exclusion bands. *Wine Industry Association of Western Australia Report* 04/07: 1–12.
- LE VIEUX, P.D. & MALAN, A.P. 2015. Prospects for using entomopathogenic nematodes to control the vine mealybug, *Planococcus ficus*, in South African vineyards. *South African Journal of Enology and Viticulture* **36**: 59–70.
DOI: [10.21548/36-1-937](https://doi.org/10.21548/36-1-937)
- LOMER, C.J., BATEMAN, R.P., JOHNSON, D.L., LANGEWALD, J. & THOMAS, M. 2001. Biological control of locusts and grasshoppers. *Annual Review of Entomology* **46**(1): 667–702.
DOI: [10.1146/annurev.ento.46.1.667](https://doi.org/10.1146/annurev.ento.46.1.667)
- LOUNSBURY, C.P. 1896. The calandra (*Phlyctinus callosus*, Bohem). *Agricultural Journal of the Cape of Good Hope* **9**: 63–64.
- LU, D., MACCHIETTO, M., CHANG, D., BARROS, M.M., BALDWIN, J., MORTAZAVI, A. 2017. Activated entomopathogenic nematode infective juveniles release lethal venom proteins. *PLOS Pathogens* **13**(4): e1006302.
DOI: <https://doi.org/10.1371/journal.ppat.1006302>
- MALAN, A.P., KNOETZE, R. & MOORE, S.D. 2011. Isolation and identification of entomopathogenic nematodes from citrus orchards in South Africa and their

- biocontrol potential against false codling moth. *Journal of Invertebrate Pathology* **108**: 115–125.
DOI: [10.1016/j.jip.2011.07.006](https://doi.org/10.1016/j.jip.2011.07.006)
- MAY, B.M. 1966. Identification of the immature forms of some common soil-inhabiting weevils, with notes on their biology. *New Zealand Journal of Agricultural Research* **9**(2): 286–316.
- MORAR-BHANA, N., CRON, G.V., GRAY, V.M. & STRAKER, C.J. 2011. The placement of South African strains of *Beauveria* in a phylogeny inferred from rDNA ITS1-5.8S-ITS2 sequences. *Journal of General and Applied Microbiology* **57**(5): 269–276.
DOI: [10.2323/jgam.57.269](https://doi.org/10.2323/jgam.57.269)
- MÜLLER, E.J. 2000. Evaluation of *Metarhizium anisopliae* mycoinsecticide as an alternative locust control measure in southern Africa. M.Sc. thesis. University of Pretoria, Pretoria, South Africa.
- NYCZEPIR, A.P., PAYNE, J.A. & HAMM, J.J. 1992. *Heterorhabditis bacteriophora*: a new parasite of pecan weevil *Curculio caryae*. *Journal of Invertebrate Pathology* **60**: 104–106.
DOI: [https://doi.org/10.1016/0022-2011\(92\)90161-V](https://doi.org/10.1016/0022-2011(92)90161-V)
- PRICE, R.E., BATEMAN, R.P., BROWN, H.D. & MÜLLER, E.J. 1997. Aerial spraying trials against brown locust (*Locustana pardalina*, Walker) nymphs in South Africa using oil-based formulations of *Metarhizium flavoviride*. *Crop Protection* **16**: 345–351.
DOI: [https://doi.org/10.1016/S0261-2194\(96\)00114-7](https://doi.org/10.1016/S0261-2194(96)00114-7)
- PRINGLE, K.L., BARNES, B.N. & BLOMEFIELD, T.L. 2015. Deciduous fruit and nut trees and olive – apple. In: Prinsloo, G.H. & Uys, V.M. (Eds) *Insects of Cultivated Plants and Natural Pastures in Southern Africa*. 350–365. Entomological Society of Southern Africa, Pretoria, South Africa.
- REHNER, S.A. & KEPLER, R.M. 2017. Species limits, phylogeography and reproductive mode in the *Metarhizium anisopliae* complex. *Journal of Invertebrate Pathology* **148**: 60–66.
DOI: [10.1016/j.jip.2017.05.008](https://doi.org/10.1016/j.jip.2017.05.008)
- RONG, I.H. & GROBBELAAR, E. 1998. South African records of associations between fungi and arthropods. *African Plant Protection* **4**(1): 43–63.
- SHAHID, A.A., RAO, A.Q., BAKHSI, A. & HUSNAIN, T. 2012. Entomopathogenic fungi as biological controllers: new insights into their virulence and pathogenicity. *Archives of Biological Science, Belgrade* **64**: 21–42.
DOI: <http://dx.doi.org/10.2298/ABS1201021S>
- SHAPIRO-ILAN, D.I. 2001a. Virulence of entomopathogenic nematodes to pecan weevil larvae *Curculio caryae* (Coleoptera: Curculionidae) in the laboratory. *Journal of Economic Entomology* **94**: 7–13.
- SHAPIRO-ILAN, D.I. 2001b. Virulence of entomopathogenic nematodes to pecan weevil adults (Coleoptera: Curculionidae). *Journal of Entomological Science* **36**: 325–328.
- SHAPIRO-ILAN, D.I., CHARLES, M.J., REILLY, C. & HOTCHKISSA, M.W. 2004. Effects of combining an entomopathogenic fungi or bacterium with entomopathogenic nematodes on mortality of *Curculio caryae* (Coleoptera: Curculionidae). *Biological Control* **30**: 119–126.
DOI: <https://doi.org/10.1016/j.biocontrol.2003.09.014>
- SHAPIRO-ILAN, D.I. & GAUGLER, R. 2002. Production technology for entomopathogenic nematodes and their bacterial symbionts. *Journal of Industrial Microbiology & Biotechnology* **28**: 137–146.
DOI: <https://doi.org/10.1038/sj/jim/7000230>
- SHAPIRO-ILAN, D.I., GOUGE, D.H. & KOPPENHOFER, A.M. 2002. Factors affecting commercial success: case studies in cotton, turf and citrus. In: Gaugler, R. (Ed.) *Entomopathogenic Nematology*. 333–356. CAB International, New York, U.S.A.
- SHAPIRO-ILAN, D.I., GARDNER, W.A., FUXA, J.R., WOOD, B.W., NGUYEN, K.B., ADAMS, B.J., HUMBER, R.A. & HALL, M.J. 2003a. Survey of entomopathogenic nematodes and fungi endemic to pecan orchards of the southeastern United States and their virulence to the pecan weevil (Coleoptera: Curculionidae). *Environmental Entomology* **32**: 187–195.
- SHAPIRO-ILAN, D.I., STUART, R. & MCCOY, C.W. 2003b. Comparison of beneficial traits among strains of the entomopathogenic nematode, *Steinernema carpocapsae*, for control of *Curculio caryae* (Coleoptera: Curculionidae). *Biological Control* **28**: 129–136.
DOI: [10.1016/S1049-9644\(03\)00030-6](https://doi.org/10.1016/S1049-9644(03)00030-6)
- SIMÕES, N. & ROSA, J.S. 1999. Pathogenicity and host specificity of entomopathogenic nematodes. *Biocontrol Science and Technology* **6**: 403–411.
- SIMSER, D. & ROBERTS, S. 1994. Suppression of strawberry root weevil, *Otiorynchus ovatus*, in cranberries by entomopathogenic nematodes (Nematoda, Steinernematidae and Heterorhabditidae). *Nematologica* **40**: 456–462.
- SMITH, M.T., GEORGIS, R., NYCZEPIR, A.P. & MILLER, R.W. 1993. Biological control of the pecan weevil, *Curculio caryae* (Coleoptera: Curculionidae), with entomopathogenic nematodes. *Journal of Nematology* **25**: 78–82.
- STUART, R.J., SHAPIRO-ILAN, D.I., JAMES, R.R., NGUYEN, K.B. & MCCOY, C.W. 2004. Virulence of new and mixed strains of the entomopathogenic nematode *Steinernema riobrave* to larvae of the citrus root weevil *Diaprepes abbreviatus*. *Biological Control* **30**(2): 439–445.
DOI: <https://doi.org/10.1016/j.biocontrol.2003.11.004>
- VERMEULEN, J.B., SWEET, S., KRAUSE, M., HOLLINGS, N. & NEL, A. 1990. *A Guide to the Use of Pesticides and Fungicides in the Republic of South Africa*. 34th revised edn. Department of Agricultural Development, Pretoria, South Africa.
- ZIMMERMANN, G. 1993. The entomopathogenic fungus *Metarhizium anisopliae* and its potential as a biocontrol agent. *Pesticide Science* **37**: 375–379.
DOI: <https://doi.org/10.1002/ps.2780370410>
- ZIMMERMANN, G. 2007a. Review on safety of the entomopathogenic fungus *Metarhizium anisopliae*. *Biocontrol Science and Technology* **17**(9): 879–920.
DOI: <https://doi.org/10.1080/09583150701593963>
- ZIMMERMANN, G. 2007b. Review on safety of the entomopathogenic fungi *Beauveria bassiana* and *Beauveria brongniartii*. *Biocontrol Science and Technology* **17**(6): 553–596.
DOI: <https://doi.org/10.1080/09583150701309006>